

Ratios and Equivalent Ratio Tables

Answer Key

Grade 6–7 Mathematics

Quick Answers

1. $\frac{2}{3}$	5. 32	9. 21
2. $\frac{5}{11}$	6. 8	10. 27
3. 12	7. $\frac{2}{9}$	
4. 4	8. —	

Common wrong answers

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4. If they got 0.25: wrote the ratio upside down (smoothies to strawberries)
 5. If they got 20: used only one of the two tape strips (5 boxes, not 8)
 7. If they got 0.29: wrote the part-to-part ratio instead of part-to-whole
 10. If they got 22.5: split the 45 L in half instead of into 3 + 2 equal parts
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1. Tile strip: ratio of red tiles to blue tiles, all three forms and simplest form.

Step 1 (read the model): count the letters — 4 R tiles and 6 B tiles. Step 2 (write the three forms): 4 to 6, 4:6, $\frac{4}{6}$. All three name the same comparison; only the notation changes.

Step 3 (simplify): divide both parts by the common factor 2, giving 2 to 3, or 2:3, or $\frac{2}{3}$.

Note: simplifying a ratio does not change the strip — it says “for every 2 red there are 3 blue.”

2. Same strip: red tiles to all the tiles.

Step 1 (find the whole): the whole handful is $4 + 6 = 10$ tiles, so this is a part-to-whole comparison, not part-to-part. Step 2 (write it): red to all is $\frac{4}{10}$. Step 3 (simplify and

interpret): $\frac{2}{5}$ — 2 out of every 5 tiles in the handful are red. Part-to-whole ratios are the ones that can also be read as a fraction of the group.

3. Ratio table: flour and water in the ratio 3 to 2; flour for 8 cups of water.

Step 1 (find the multiplier): the water row goes 2, 4, 6, 8 — each column is the first column times 1, 2, 3, 4. Step 2 (scale both rows the same way): flour goes 3, 6, 9, 12. Equivalent ratios come from multiplying *both* quantities by the same number. Step 3 (read the asked column): $\frac{x}{8} = \frac{3}{2}$ gives $2x = 24$, so $x = \boxed{12}$ cups of flour.

4. Value of the ratio 12 strawberries to 3 smoothies.

Step 1 (write the ratio in the asked order): strawberries to smoothies is 12 to 3, or $\frac{12}{3}$. Step 2 (find the value): the value of a ratio is the quotient of its two numbers, $12 \div 3$. Step 3: $\boxed{4}$ strawberries per smoothie. Writing the ratio the other way, $\frac{3}{12}$, answers a different question (smoothies per strawberry).

5. Tape diagram: cats to dogs is 3 to 5, each box is 4 animals; total animals.

Step 1 (count boxes): 3 boxes of cats and 5 boxes of dogs, so the diagram has $3 + 5 = 8$ equal boxes in all. Step 2 (use the box value): every box is worth the same 4 animals, so the total is 8×4 . Step 3: $\boxed{32}$ animals (12 cats and 20 dogs, which checks: $12 : 20$ simplifies back to $3 : 5$).

6. Ratio table: 5 pencils cost 2 dollars; cost of 20 pencils.

Step 1 (find the multiplier): from 5 pencils to 20 pencils is a factor of 4. Step 2 (scale both rows): cost goes 2×4 ; the middle column (10 pencils) is $2 \times 2 = 4$ dollars, which is a useful check. Step 3: $\frac{x}{20} = \frac{2}{5}$ gives $5x = 40$, so the bundle costs $\boxed{8}$ dollars.

7. Part to whole from a $2 : 7$ part-to-part ratio.

Step 1 (name the parts): the ratio $2 : 7$ compares walkers with riders — both are parts, so 7 is not the whole. Step 2 (build the whole): the class is $2 + 7 = 9$ equal shares. Step 3 (write the fraction): walkers are 2 of those 9 shares, so the fraction of the class that walks is $\boxed{\frac{2}{9}}$. That is where the 9 comes from: it is $2 + 7$, the total number of shares.

8. Compare Mix A (4 scoops in 10 cups) with Mix B (6 scoops in 14 cups).

Step 1 (write both as fractions): Mix A is $\frac{4}{10}$ and Mix B is $\frac{6}{14}$. Step 2 (find each value): $4 \div 10 = 0.4$ scoops per cup; $6 \div 14 = 0.428 \dots$ scoops per cup. (Common denominators work too: $\frac{4}{10} = \frac{28}{70}$ and $\frac{6}{14} = \frac{30}{70}$.) Step 3 (compare in the asked order): 0.4 is less than $0.428 \dots$, so value of Mix A $\boxed{<}$ value of Mix B — Mix B is the stronger mix.

9. Ratio table: 9 lemons per 6 cups; lemons for 14 cups.

Step 1 (take the easy step first): divide both quantities by 3 to reach the 2-cup column — $6 \text{ cups} \div 3 = 2 \text{ cups}$ and $9 \text{ lemons} \div 3 = 3 \text{ lemons}$. Step 2 (scale up from there): 14 cups is 7 times 2 cups, so multiply both by 7: $3 \times 7 = 21$ lemons. Step 3 (check with the proportion): $\frac{x}{14} = \frac{9}{6}$ gives $6x = 126$, so $x = \boxed{21}$ lemons.

10. Synthesis: blue to yellow is 3 to 2, and 45 liters of mixed paint are needed; liters of blue.

Step 1 (model the whole): the mix is $3 + 2 = 5$ equal boxes of paint, so the part-to-whole ratio for blue is $\frac{3}{5}$. Step 2 (find one box): the 5 boxes together are 45 liters, so one box is $45 \div 5 = 9$ liters. Step 3 (take the blue boxes): blue is 3 boxes, $3 \times 9 = 27$, so Nia needs $\boxed{27 \text{ L}}$ of blue (and 18 L of yellow, which totals 45 L).